Operations Research

Linear Programming II: Solving linear programs

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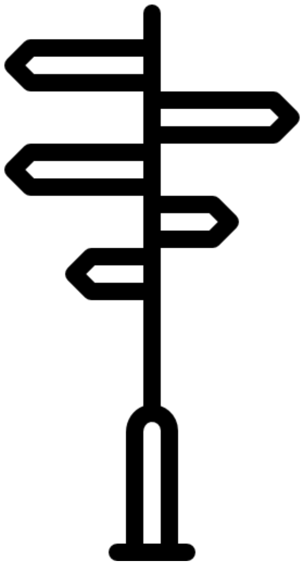
Course outline



- **Linear Optimization**
  - Modeling with linear programming and graphical solution
  - **Solving linear programs, convexity**
  - Simplex algorithm, 2-phase Simplex
  - Simplex algorithm in matrix notation, goal programming
  - Sensitivity analysis, reoptimization
  - Duality theory, zero-sum games

- **Integer Linear Optimization**
  - Modeling integer programming problems (IPs)
  - Solving IPs: branch & bound, cutting planes
  - Advanced solution methods: column generation, approximation algorithms
  - Fast solvable integer problems: total unimodularity, matroids
  - Network flow problems
  - Traveling salesperson problem

- **Nonlinear Optimization**
  - Optimization of multidimensional functions
  - Convex optimization

Today you will learn ...

- to analyze the structure of LPs and their solution **geometrically und algebraically**.

- to describe the **basics of the Simplex Algorithm**.

Techniques for solving LPs

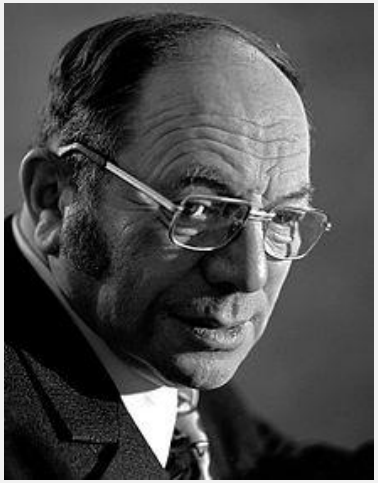
- Real-valued LPs
  - Simplex
  - Interior-point method
  - Ellipsoid method
  - Numerous special algorithms for specific problems (e.g. transportation or assignment problems)

Lectures 11 - 12

- Integer LPs (IP)
  - Branch-and-Bound
  - Cutting-plane
  - Branch-and-Cut
  - Meta-Heuristics (no exact solution guaranteed)

Lectures 8 - 10

Leonid Kantorovich



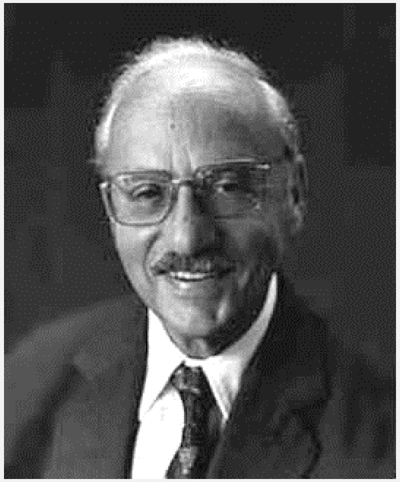
- born on 19.01.1912 in St. Petersburg, died on 07.04.1986 in Moscow
- “Mathematical Methods of Organizing and Planning Production“ (1939)
- established linear optimization

Nobel Memorial Prize in Economic Sciences 1975

1975		Leonid Kantorovich (1912–1986)	Soviet Union	"for their contributions to the theory of optimum allocation of resources"[13]	Leningrad State University	Novosibirsk State University	Linear programming, Kantorovich theorem, Kantorovich inequality, Kantorovich metric
		Tjalling Koopmans (1910–1985)	Netherlands United States		University of Leiden	University of Chicago Yale University	Linear programming

source: en.wikipedia.org

George Dantzig



- born on 08.11.1914 in Portland, died on 13.05.2005
- "Simplex Method of Optimization“ (1947),
- originated from work at the USAF
- further developments by John von Neumann and Oskar Morgenstern

“*It is interesting to note that the original problem that started my research is still outstanding - namely the problem of planning or scheduling dynamically over time, particularly planning dynamically under uncertainty. If such a problem could be successfully solved it could eventually, through better planning, contribute to the well-being and stability of the world*.”

Why study LP algorithms?

Important for implementing advanced methods and for fully exploiting the solver:

- cutting / branching in integer programming
- many variants also in the Simplex Algorithm
- column generation / pricing

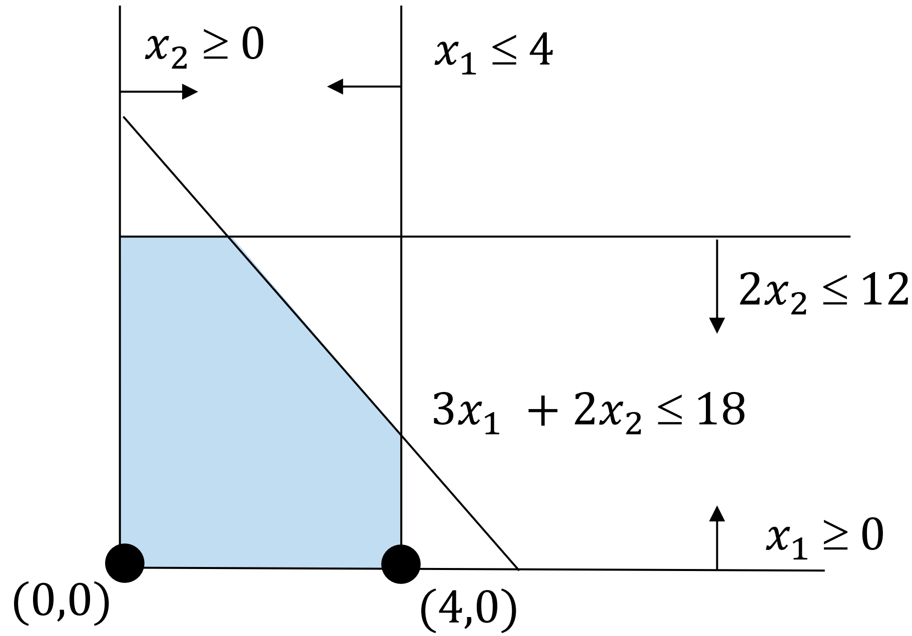
Sensitivity analysis

Bounds / Duality / Shadow prices

Relaxations

LP: Graphical representation

$$
\begin{array}{llcccc}
\text{max} & 5x_1 & + & 3x_2 & & \\
\text{s.t.} & x_1 & & & \leq & 4 \\
 & & & 2x_2 & \leq & 12 \\
 & 3x_1 & + & 2x_2 & \leq & 18 \\
 & & & x_1, x_2 & \geq & 0
\end{array}
$$

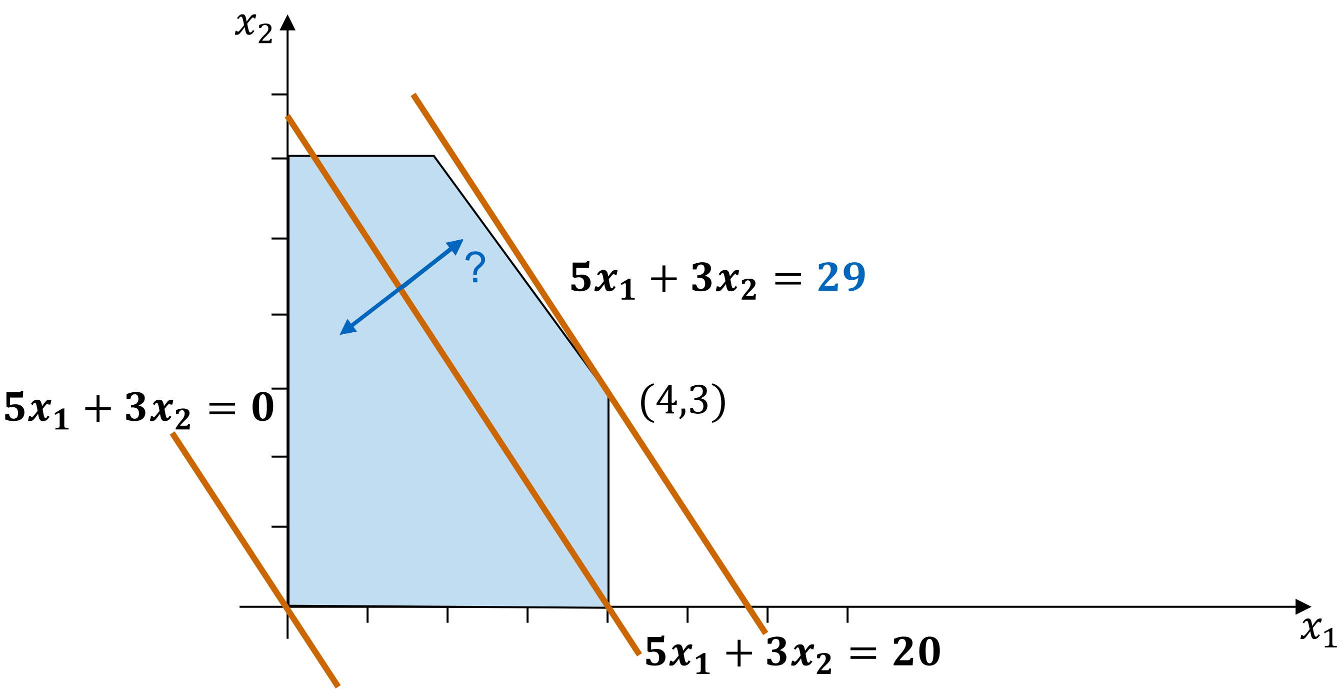


$x_1 = 0, x_2 = 0$ is a feasible solution with an objective value of 0.

$x_1 = 4, x_2 = 0$ is a feasible solution with an objective value of 20.

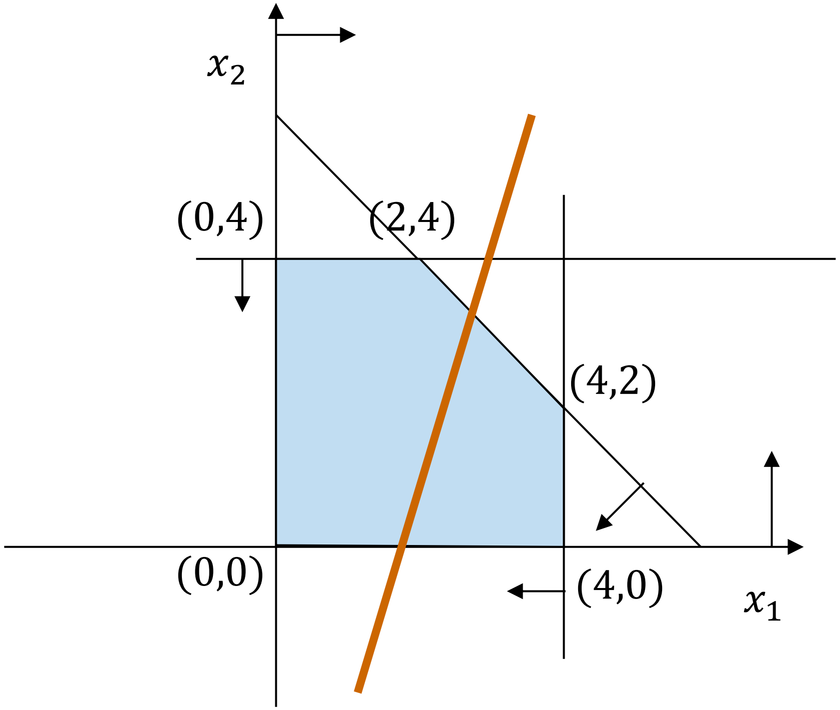
LP: Graphical solution

$(4{,}0)$ has objective value 20.
All other solutions with objective value 20 lie on $5x_1 + 3x_2 = 20$



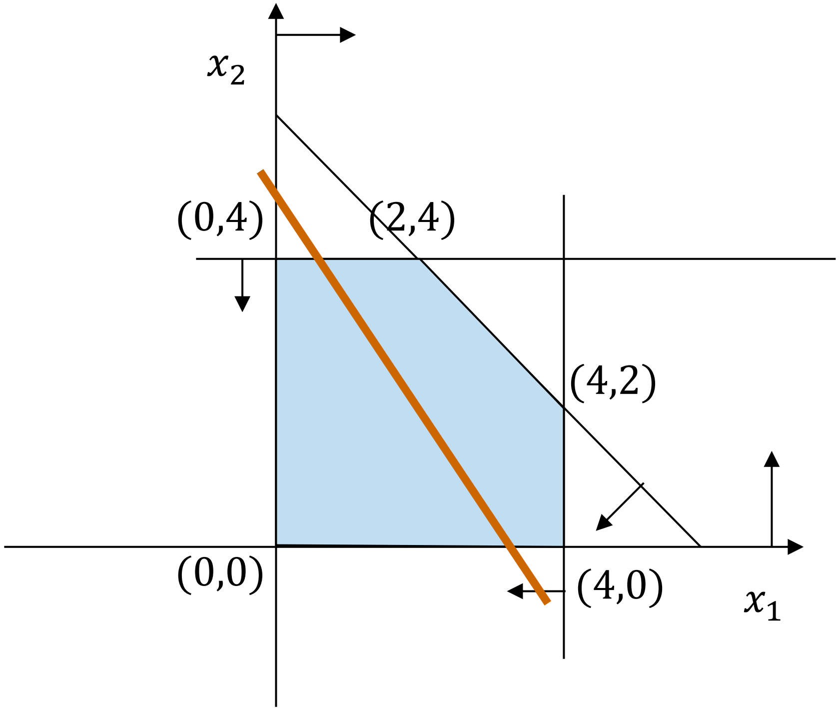
Find the optimal solution

$$
\begin{array}{lcccccc}
\text{max} & 3x_1 & - & x_2 & & \\
\text{s.t.} & x_1 & + & x_2 & \leq & 6 \\
 & x_1 & & & \leq & 4 \\
 & & & x_2 & \leq & 4 \\
 & & & x_1, x_2 & \geq & 0
\end{array}
$$



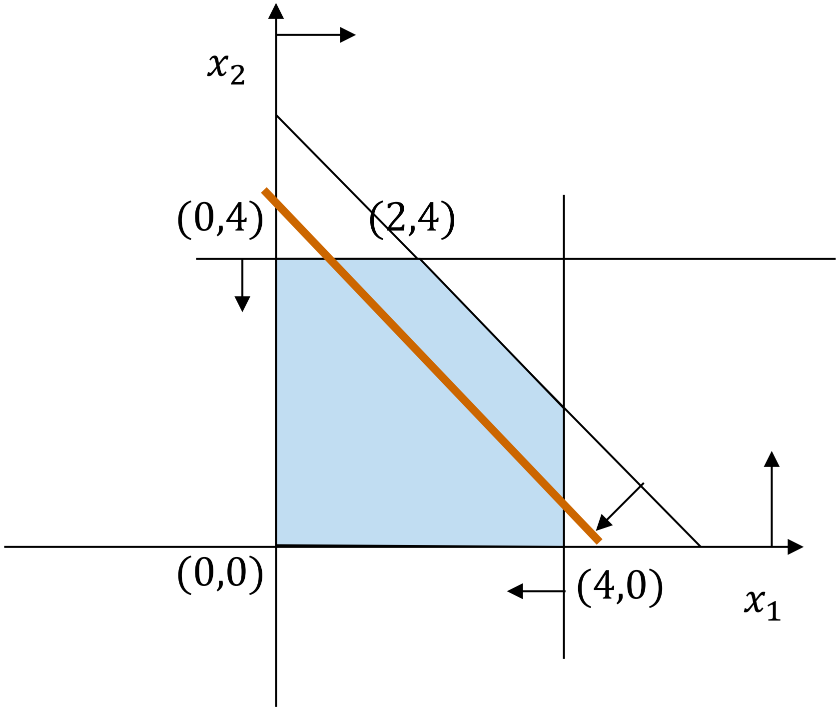
Find the optimal solution

$$\begin{array}{lllllll}
\text{max} & 2x_1 & + & x_2 & & \\
\text{s.t.} & x_1 & + & x_2 & \leq & 6 \\
 & x_1 & & & \leq & 4 \\
 & & & x_2 & \leq & 4 \\
 & & & x_1, x_2 & \geq & 0
\end{array}$$



Find the optimal solution

max	x_1	$+$	x_2		
s.t.	x_1	$+$	x_2	$\leq$	6
	x_1			$\leq$	4
			x_2	$\leq$	4
			x_1, x_2	$\geq$	0



Observation

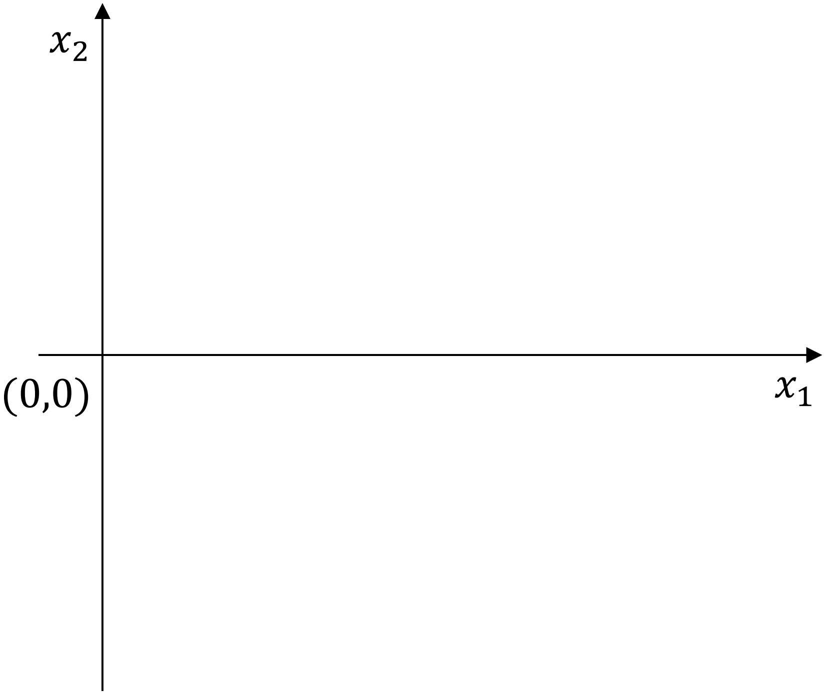
We always find an optimal solution at the edge of the feasible region. In particular, a solution always lies in one of the vertices (also called extreme points).

Naive approach

- identify all vertices of the feasible region
- select the best solution

No vertices

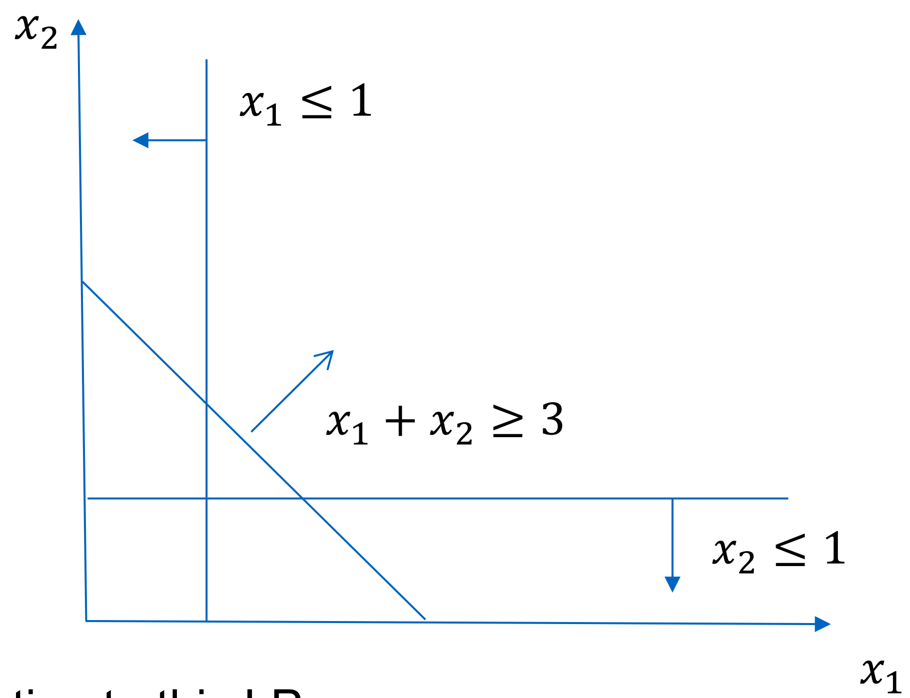
min	x_1		
s.t.	x_1	$\geq$	0
	x_2	$\in$	$\mathbb{R}$



Each solution with $x_1 = 0$ is optimal!

Unsolvability / infeasibility

$$\begin{array}{llllll}
\min & x_1 & & & & \\
\text{s.t.} & x_1 & + & x_2 & \geq & 3 \\
 & x_1 & & & \leq & 1 \\
 & & & x_2 & \leq & 1 \\
 & & & x_1, x_2 & \geq & 0
\end{array}$$

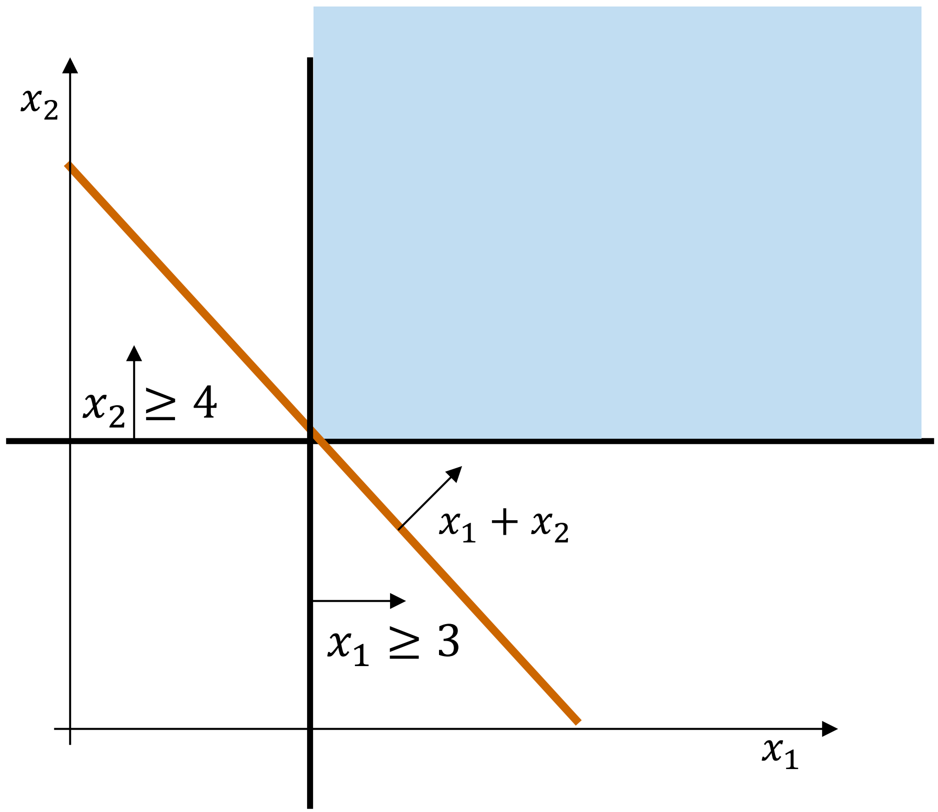


There exists no possible (feasible) solution to this LP.

There are no real numbers x_1 and x_2 such that, if both are at most 1, their sum is at least 3.

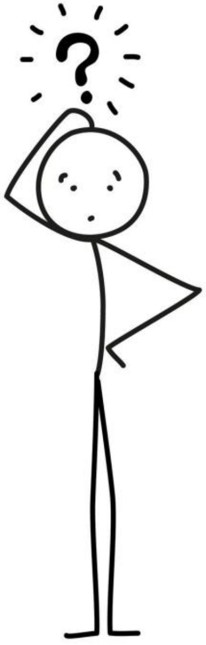
Unbounded feasible region

$$
\begin{array}{llllll}
\max & x_1 & + & x_2 & & \\
\text{s.t.} & x_1 & & & \geq & 3 \\
 & & & x_2 & \geq & 4 \\
 & & & x_1, x_2 & \geq & 0
\end{array}
$$



The LP is feasible (solvable), but the objective value is not bounded.

What solutions are possible for a linear programming problem?



Possible structure of an LP

- has one optimal solution
- has multiple optimal solutions
- infeasible (empty feasible region)
- unbounded feasible region
- the feasible region has no vertices

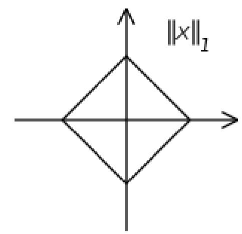
We will show the following:
If the feasible region is not empty and bounded, the optimal solutions lie at the vertices.

Inner product and norms

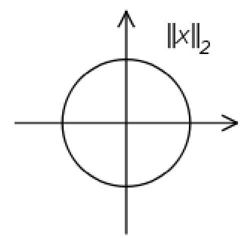
Definition (Scalar product, Inner product): $\langle x, y \rangle = x^T y = \sum_{i=1}^{n} x_i y_i \; \forall x, y \in \mathbb{R}^n$

Definition (Euclidean Norm or 2-Norm):

$$\|x\|_2 = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2} \quad \forall x = (x_1, \cdots, x_n)^T \in \mathbb{R}^n$$



The diamond is the set of all points (x,y) where the L1 norm equals 1



the circle are all points (x,y) where $\sqrt{x^2 + y^2} = 1$

Definition (Infinity Norm): $\|x\|_\infty = \max\{|x_1|, |x_2|, \cdots, |x_n|\}$

Example:

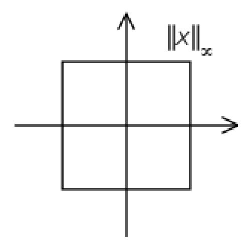
The 1-, 2-, 3- or ∞ –norms of the vector x=(3, -2, 6) are

$\|x\|_1 = |3| + |-2| + |6| = 11$

$\|x\|_2 = \sqrt{|3|^2 + |-2|^2 + |6|^2} = \sqrt{49} = 7$

$\|x\|_3 = \sqrt[3]{|3|^3 + |-2|^3 + |6|^3} = \sqrt[3]{251} = 6{,}308$

$\|x\|_\infty = \max\{\,|3|, |-2|, |6|\} = 6$



Norms, metrics, and vector spaces

Informal Definition (norm): A **norm** assigns a non-negative length or size to each vector.

Informal Definition (metric): A **metric** or distance function on a set X is a function that defines the distance between every pair of elements in X. It is denoted as $d(x, y)$ for elements $x, y \in X$.

Informal Definition (vector space): A **vector space** is a collection of vectors, which are objects that can be added together and multiplied ("scaled") by numbers, known as scalars.

Definition (Euclidean space): $\mathbb{R}^n$ is a vector space, consisting of all ordered *n*-tuples of real numbers (where *n* is a positive integer). It is equipped with the Euclidean metric, which allows for measuring distances and angles:

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

where *x* and *y* are points in $\mathbb{R}^n$.

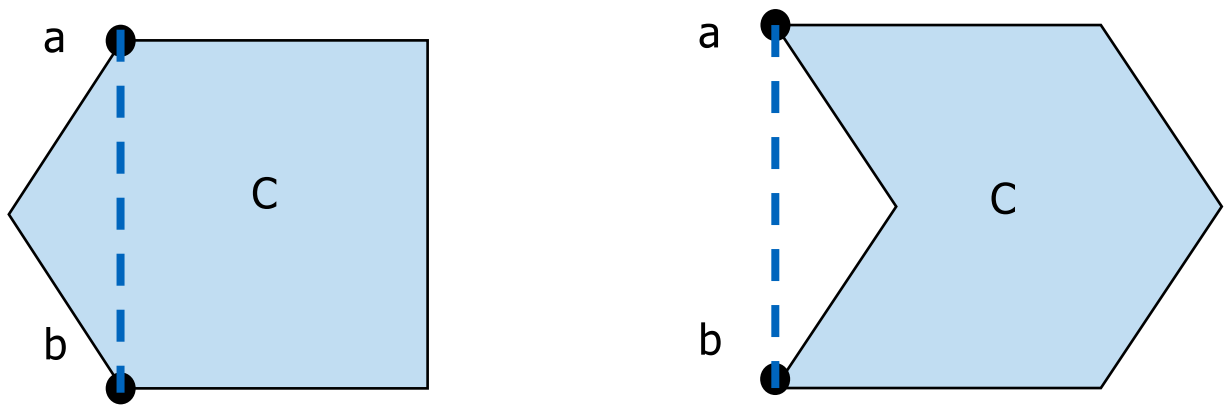
Convex sets

Definition (Convex Set):
A set $C \subset \mathbb{R}^n$ is convex if for $x, y \in C$ the connecting line

$$[x, y] \coloneqq \{(1-\lambda)x + \lambda y | 0 \le \lambda \le 1\}$$

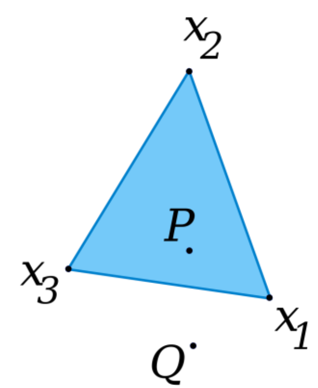
lies in C.

The **convex hull** is the smallest convex set that includes C.



Definitions from Linear Algebra

Definition (Convex Combination):
A convex combination of k vectors $(x_1, \cdots, x_k)$ is a linear combination of the vectors where all coefficients are non-negative and sum to 1.
$\sum_{i=1}^{k} \lambda_i x_i$ with $\sum_{i=1}^{k} \lambda_i = 1$ and $\lambda_i \geq 0, i = 1, \cdots, k$.



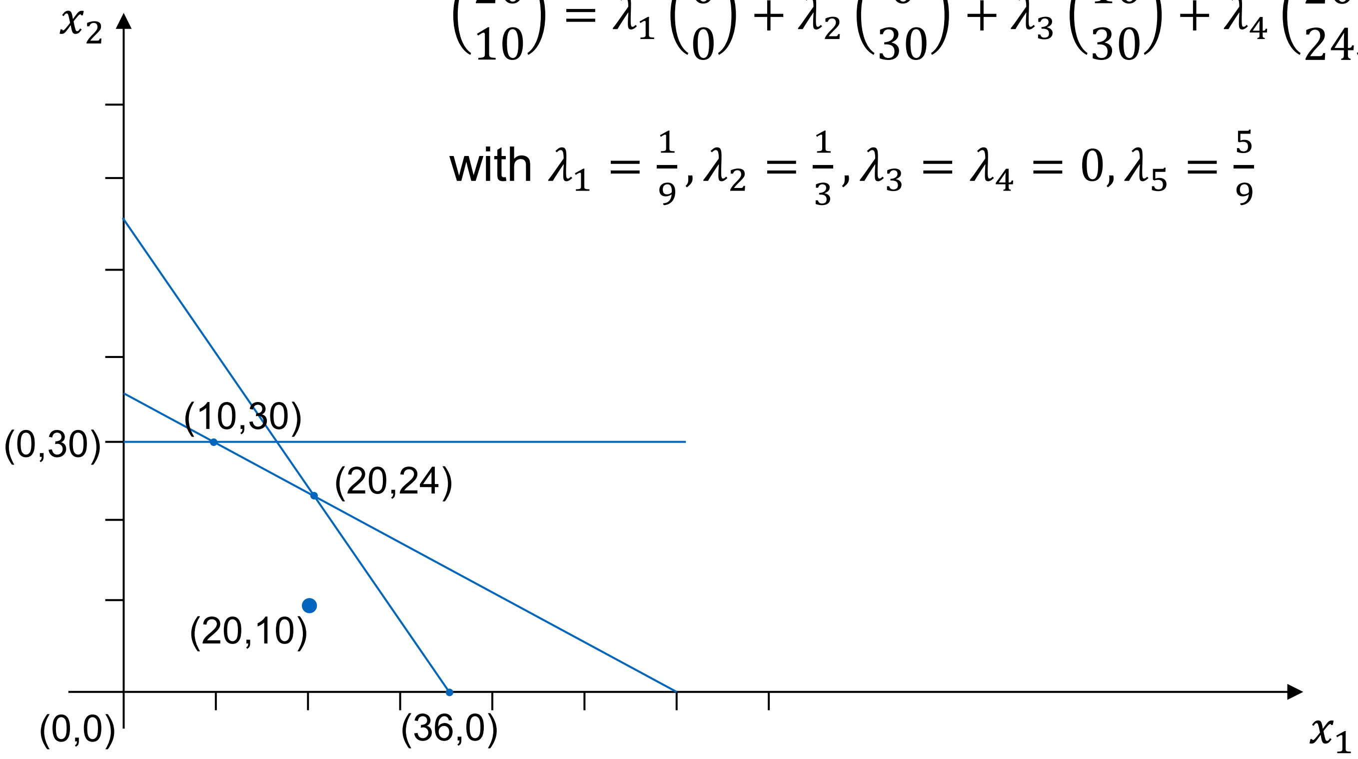
Theorem:
A set $C \subset \mathbb{R}^n$ is convex iff it includes all convex combinations of the points in C.

Definition (Ball):
For $x \in \mathbb{R}^n, r > 0$ and an arbitrary norm $||\cdot||$ on $\mathbb{R}^n$, the *open ball* around x with radius r is
$B(x,r) = \{y \in \mathbb{R}^n | \; ||x - y|| < r\}$, and the *closed ball* around x with radius r is
$B(x,r) = \{y \in \mathbb{R}^n | \; ||x - y|| \leq r\}$.

Convex combination

$$\binom{20}{10} = \lambda_1 \binom{0}{0} + \lambda_2 \binom{0}{30} + \lambda_3 \binom{10}{30} + \lambda_4 \binom{20}{24} + \lambda_5 \binom{36}{0}$$

with $\lambda_1 = \frac{1}{9}, \lambda_2 = \frac{1}{3}, \lambda_3 = \lambda_4 = 0, \lambda_5 = \frac{5}{9}$



LP and convexity

Theorem 1:
The feasible region $\mathrm{P} = \{x \in \mathbb{R} | Ax = b\ , x \geq 0\ \}$ of a linear program is convex.

<u>Proof:</u>
Let $y, z \in P$ and $x = \lambda y + (1 - \lambda)z$ with $0 \leq \lambda \leq 1$. Then
- $x \geq 0$
- $Ax = \lambda Ay + (1 - \lambda)Az = \lambda b + (1 - \lambda)b = b$

Theorem 2:
The feasible region $\mathrm{P} = \{x \in \mathbb{R} | Ax \leq b\ , x \geq 0\ \}$ of a linear program is convex.

Theorem 3:
The set of optimal solutions $\mathrm{L} = \{x \in \mathbb{R} | c^T x = Z\ , x \in P\}$ of a linear program is convex.

Convex combinations and vertices

Let $x_1, x_2, \dots, x_r$ be points in $\mathbb{R}^n$ and $\lambda_1, \lambda_2, \dots, \lambda_r$ be non-negative scalars.

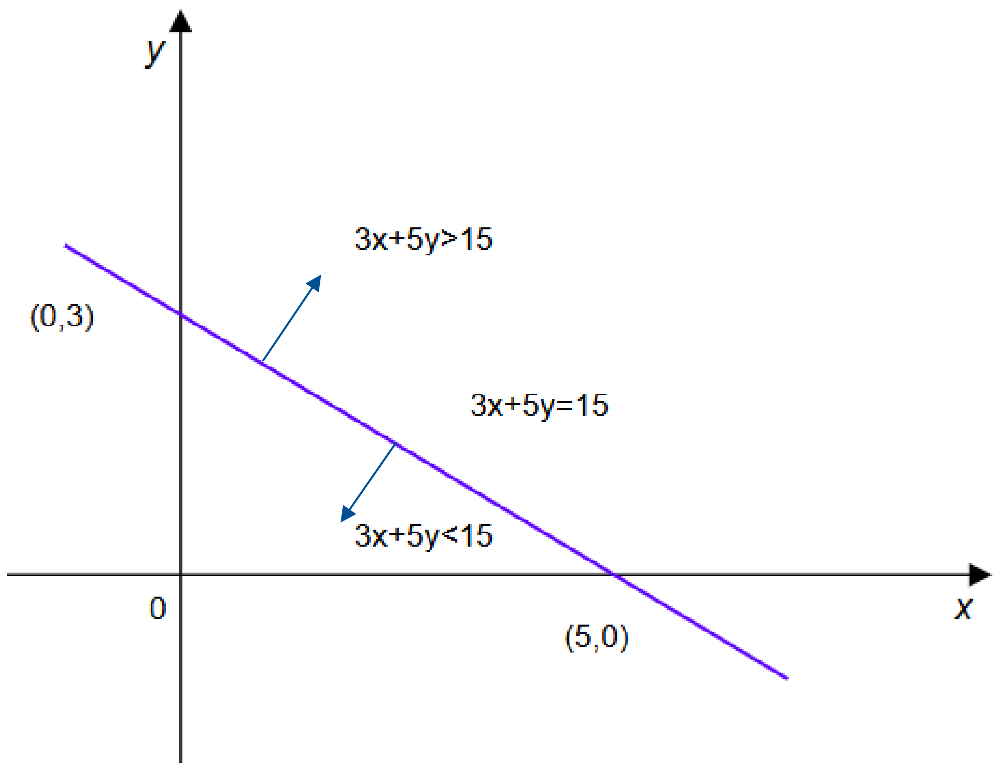
If $\sum_{i=1}^{r} \lambda_i = 1$, then $y \coloneqq \sum_{i=1}^{r} \lambda_i x_i$ is a **convex linear combination** or **convex combination** of $x_1, x_2, \dots, x_r$.

If $0 < \lambda_i < 1$ for all $i = 1, \dots, r$, then y is a **strict convex combination**.

A point x from a convex set is an **extreme point (vertex)**, if x cannot be expressed as a strict convex combination of points from this set.

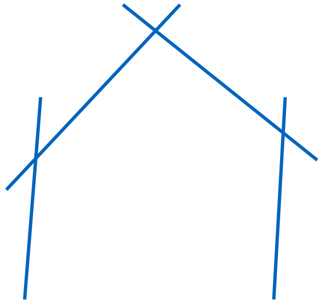
Hyperplanes and half-spaces

Hyperplane: $H = \{x: \langle a, x \rangle = b\}$
Half-space: $H^+ = \{x: \langle a, x \rangle \leq b\}, H^- = \{x: \langle a, x \rangle \geq b\}$

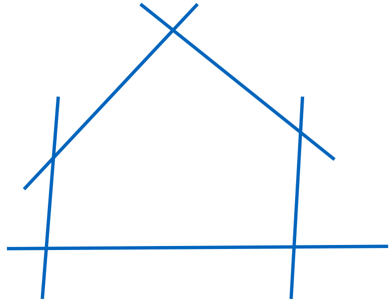


Polytope and polyhedron

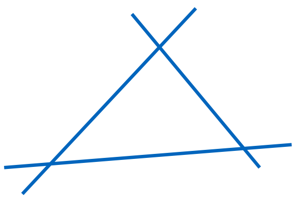
- A **half-space** is a subset of a space of arbitrary dimension that is bounded by a hyperplane.
- The intersection of finitely many half-spaces $\{x \in \mathbb{R}^n | a_j x \leq b_j\}$ is called **polyhedron**.
  – polyhedra are not necessarily bounded.
- A bounded, non-empty polyhedron is called **polytope**.
- The set of all strict convex combinations of finitely many points is a **convex polytope** (=convex hull of the points).
- A convex polytope which is spanned by $n + 1$ points in $\mathbb{R}^n$ that do not lie on a hyperplane is called **simplex**.



convex polyhedron



convex polytope

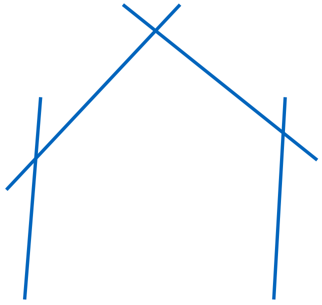


simplex

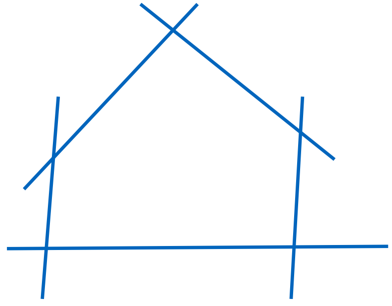
Polytope and polyhedron

Theorem:
The feasible set $\mathrm{P} = \{x \in \mathbb{R} | Ax \leq b \, , x \geq 0 \, \}$ of a linear optimization problem is a convex polyhedron. If it is bounded, it is a polytope defined by the intersections of the bounding hyperplanes.

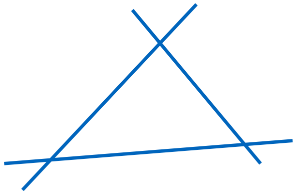
Rationale: Intersection of convex half-spaces is convex.



convex polyhedron



convex polytope



simplex

Optimal solutions

Theorem:
If a linear optimization problem has an optimal solution, then at least one vertex of the feasible region is an optimal solution.

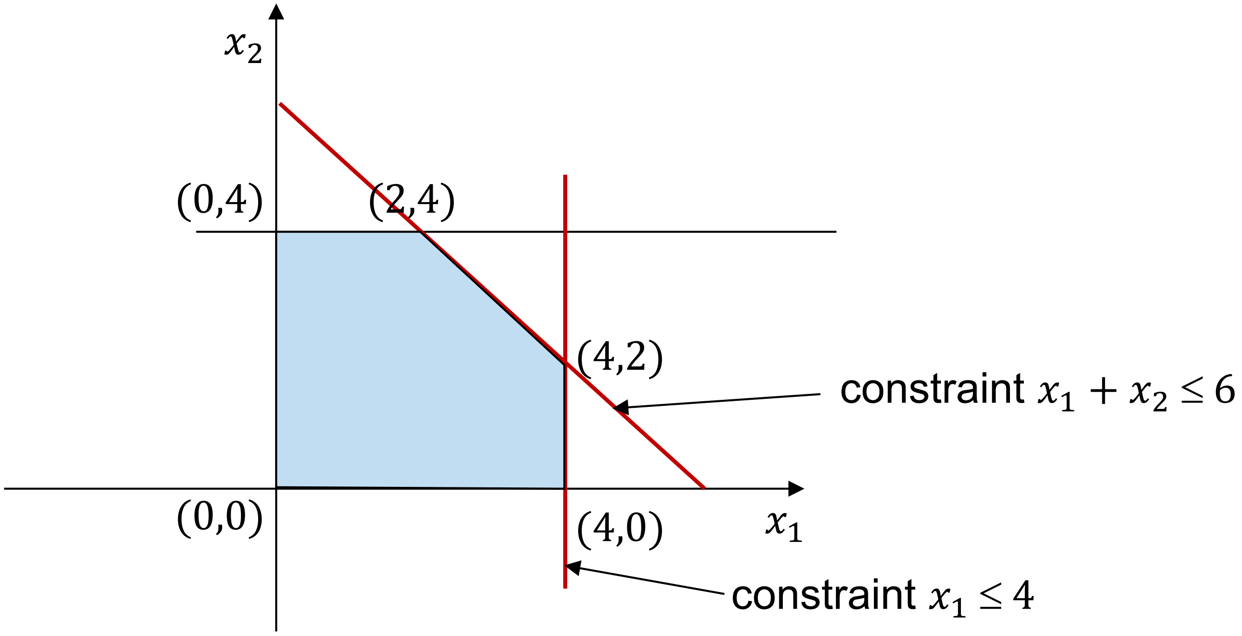
This is suggesting the following approach:
The optimal solution of an LP can be found by enumerating all vertices of the feasible region.

Previous method: graphical representation.

How to find vertices in multidimensional problems with $n > 2$?

Vertices of the LP

In the 2-dimensional case, each vertex lies at the intersection of 2 straight lines (constraints).



For an LP with n variables, each vertex solution lies at the intersection of n hyperplanes (constraints).

Generalization

2 variables

feasible region: $\mathbb{R}^2$

bounded by straight lines

solution at intersection of
2 straight lines

$\Rightarrow$

n variables

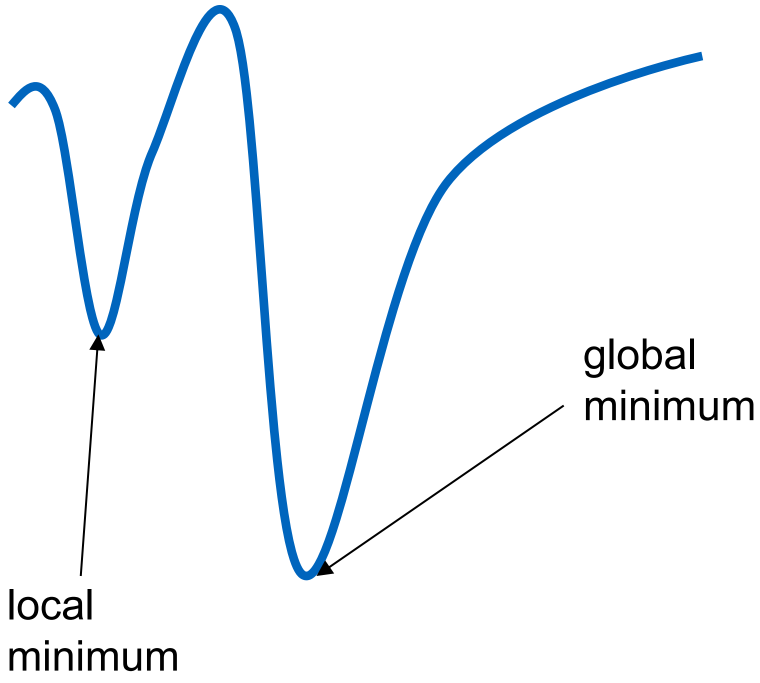
feasible region: $\mathbb{R}^n$

bounded by hyperplanes

solution at intersection of
n Hyperplanes

Finding the optimal solution

Theorem:
A local minimum of the objective function of a linear optimization problem on the feasible region is also a global minimum.



Adjacent vertices

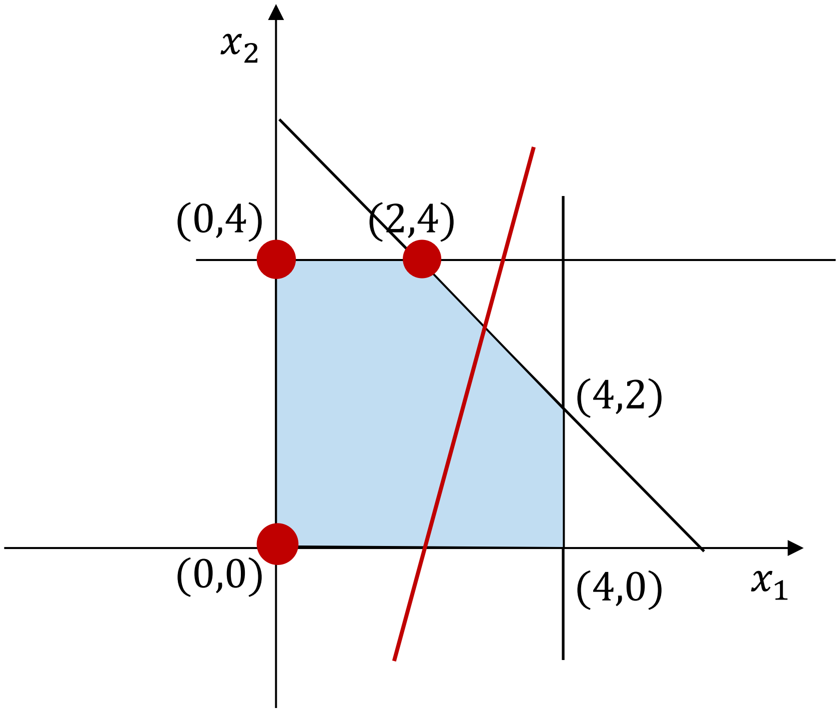
Theorem:
A local minimum of the objective function of a linear optimization problem on the feasible region is also a global minimum.

→ Therefore, to check the optimality of a vertex, we need to check only the adjacent vertices and not all of them.

- Two vertices are adjacent if they share n-1 restrictions (one "goes across exactly one hyperplane to the neighboring vertex").
- Given a vertex, if there is no adjacent vertex with a better objective value, then this vertex is optimal.
- We can iterate over the vertices one by one to find the optimum.
  - In the worst case we need to consider all points (rarely the case).

Example

$$\begin{array}{llllll} \text{min} & 3x_1 & - & x_2 & & \\ \text{s.t.} & x_1 & + & x_2 & \leq & 6 \\ & x_1 & & & \leq & 4 \\ & & & x_2 & \leq & 4 \\ & & & x_1, x_2 & \geq & 0 \end{array}$$



The objective value at point (0,4) is -4.
The adjacent vertices are (0,0) and (2,4).
The objective value at point (0,0) is 0.
The objective value at point (2,4) is 2.
↳ (0,4) is an optimal solution.

Observations

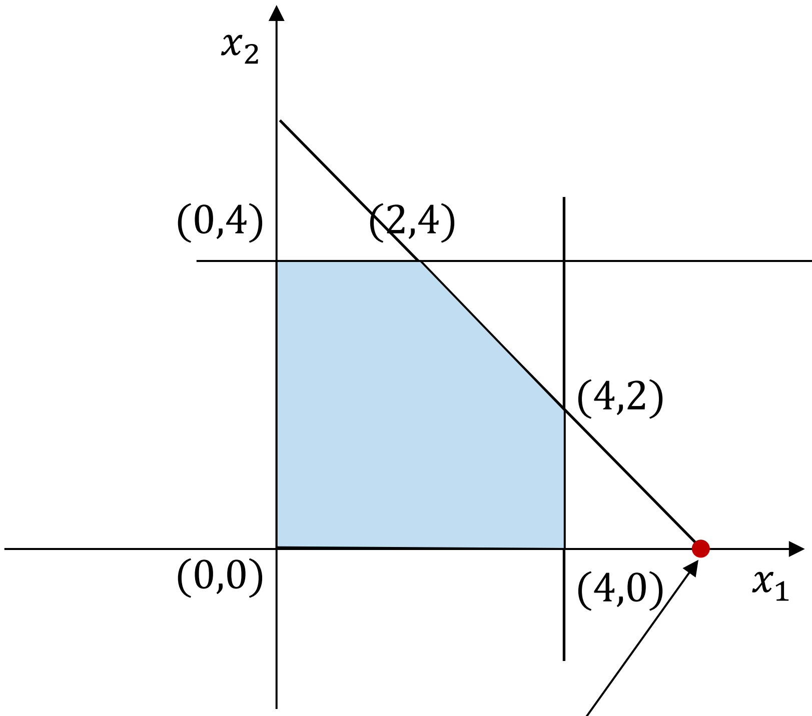
The intersection of constraints is obtained by solving linear equations.

Can we find all vertices by finding the intersection of each n constraints?
- Not each intersection of n constraints is also a vertex of the feasible region.

Some vertices may lie at the intersection of more than n constraints.
- But every possible vertex solution lies at the intersection of at least n constraints.

Example I

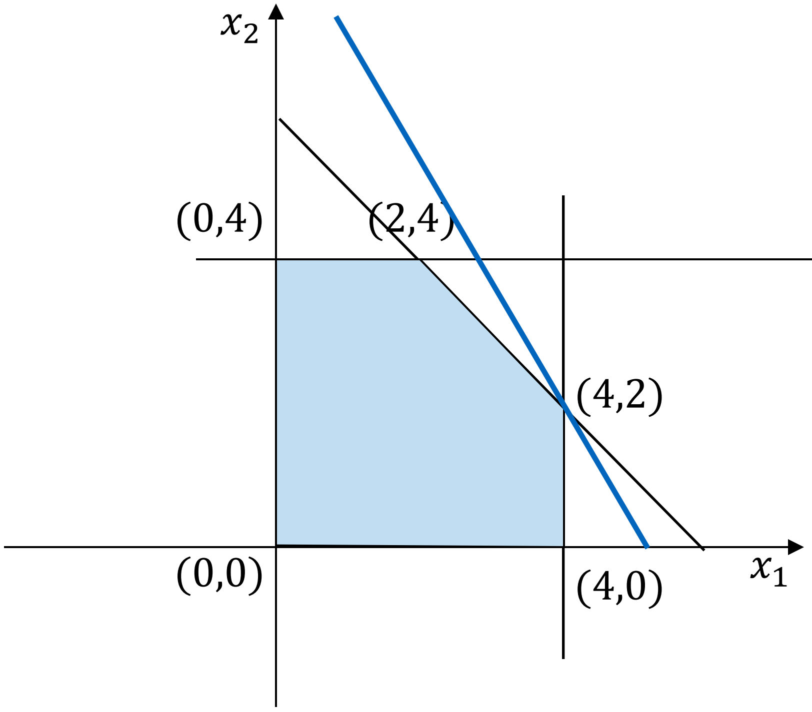
$$
\begin{array}{llllll}
\text{max} & 2x_1 & + & x_2 & & \\
\text{s.t.} & x_1 & + & x_2 & \leq & 6 \\
 & x_1 & & & \leq & 4 \\
 & & & x_2 & \leq & 4 \\
 & & & x_1, x_2 & \geq & 0
\end{array}
$$



Intersection between
$x_1 + x_2 \leq 6$ and $x_2 \geq 0$

Example II

$$
\begin{array}{llllll}
\max & 2x_1 & + & x_2 & & \\
\text{s.t.} & x_1 & + & x_2 & \leq & 6 \\
 & x_1 & & & \leq & 4 \\
 & & & x_2 & \leq & 4 \\
 & & & x_1, x_2 & \geq & 0
\end{array}
$$



Additional constraint:

$x_1 + \frac{1}{2} x_2 \leq 5$

This constraint is a linear combination of $x_1 \leq 4$ and $x_1 + x_2 \leq 6$!

Implementation issues

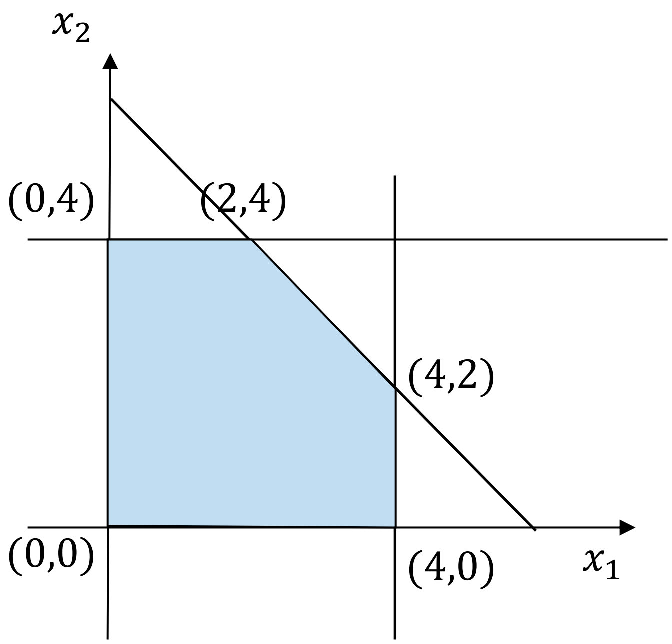
- How to find the (feasible) vertices?
- How to identify adjacent vertices?
  - Adjacent vertices share n-1 constraints.
  - Thus, we can enumerate all n constraints to compute all adjacent vertices that share at least n-1 constraints with the currently considered vertex.

Implementation issues

- **How to find the (feasible) vertices?**
- **How to identify adjacent vertices?**
  - **Adjacent vertices share n-1 constraints.**
  - **Thus, we can enumerate all n constraints to compute all adjacent vertices that share at least n-1 constraints with the currently considered vertex.**

Example

max	$2x_1$	$+$	x_2			
s.t.	x_1	$+$	x_2	$\leq$	6
	x_1			$\leq$	4
			x_2	$\leq$	4
			x_1, x_2	$\geq$	0



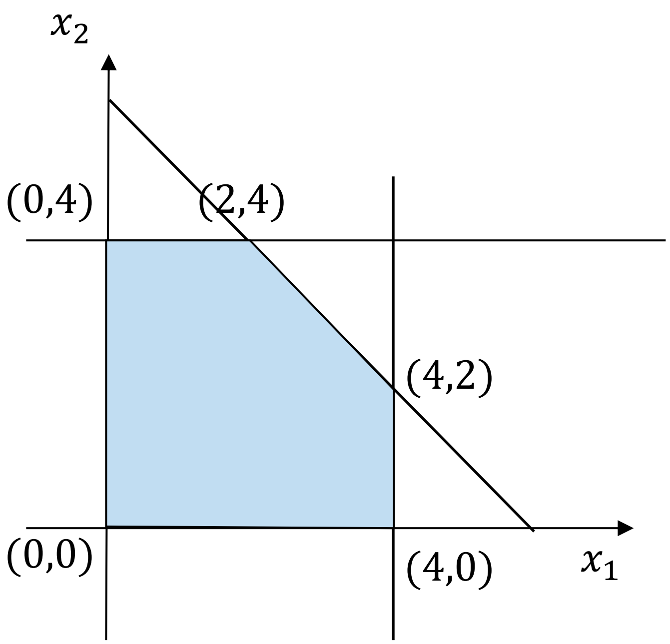
(0,0) satisfies constraints $x_1 \geq 0$ and $x_2 \geq 0$.
Sets of 2 constraints, of which one is either $x_1 \geq 0$ or $x_2 \geq 0$:

Set	**Point**
$x_1 + x_2 \leq 6, x_1 \geq 0$	$(0,6)$ (not allowed)
$x_1 \leq 4, x_1 \geq 0$	no intersection point
$x_2 \leq 4, x_1 \geq 0$	$(0,4)$
$x_1 + x_2 \leq 6,\ x_2 \geq 0$	$(6,0)$ (not allowed)
$x_1 \leq 4,\ x_2 \geq 0$	$(4,0)$
$x_2 \leq 4,\ x_2 \geq 0$	no intersection point

Finding vertices

- Vertices are obtained by solving linear equations.
- If the LP is given in equation form, we can determine the vertices as the solutions of the linear system of equations.
- So far, we have represented LPs using inequalities (the constraints of the feasible polytope).

$$\begin{array}{llcllll} \text{max} & 2x_1 & + & x_2 & & \\ \text{s.t.} & x_1 & + & x_2 & \leq & 6 \\ & x_1 & & & \leq & 4 \\ & & & x_2 & \leq & 4 \\ & & & x_1, x_2 & \geq & 0 \end{array}$$

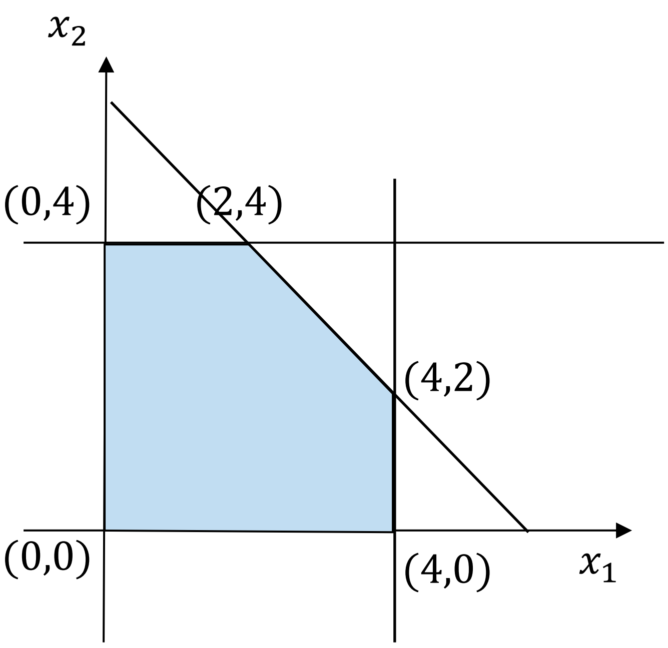


Finding vertices

- Vertices are obtained by solving linear equations.
- If the LP is given in equation form, we can determine the vertices as the solutions of the linear system of equations.
- So far we have represented LPs using inequalities (the constraints of the feasible polytope).

Conversion to **standard form** by introducing **slack variables**:

$$
\begin{array}{llllllll}
\text{max} & 2x_1 & + & x_2 & & & & \\
\text{s.t.} & x_1 & + & x_2 & +\boldsymbol{x_3} & & & = 6 \\
& x_1 & & & & +\boldsymbol{x_4} & & = 4 \\
& & & x_2 & & & +\boldsymbol{x_5} & = 4 \\
& & & & & & x_1, x_2, \boldsymbol{x_3}, \boldsymbol{x_4}, \boldsymbol{x_5} & \geq 0
\end{array}
$$



Standard form, normal form, canonical form

Normal form:

$$\begin{aligned} &\max\ c^T x \\ &\text{s.t.}\quad Ax \le b \\ &\qquad\quad x \ge \vec{0} \end{aligned} \qquad\qquad \begin{aligned} &\min\ c^T x \\ &\text{s.t.}\quad Ax \ge b \\ &\qquad\quad x \ge \vec{0} \end{aligned}$$

Standard form (with $b \geq \vec{0}$)**:**

$$\begin{aligned} &\max\ \underline{c^T x} \\ &\text{s.t.}\quad Ax = b \\ &\qquad\quad x \ge \vec{0} \end{aligned}$$

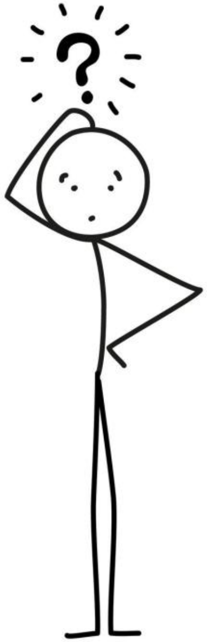
Be aware! The terms

- standard form
- normal form
- canonical form *(later!)*

are used inconsistently in the literature.
We use Winston's notation in this lecture.

The simplex algorithm requires an LP in standard form.
For formulating and analyzing the dual problem, the normal form is very helpful.

1. Why is a linear program convex?
2. How do normal form and standard form differ?



Polyhedron in standard form

<u>Standard form</u> (with $b \geq \vec{0}$)**:** max $\quad c^T x$

$$\begin{aligned} \text{s.t.} \quad Ax &= b \\ x &\geq \vec{0} \end{aligned}$$

Theorem:
Every possible vertex solution for an LP in standard form with n variables and m constraints has at least n-m variables equal to 0.

Proof: Suppose a point in the feasible space had more than m positive variables. Such a point can be described as the solution of a system of linear equations. With more than m columns, these columns would be linearly dependent.
So one can move in a nonzero null direction while keeping feasibility. Hence the point cannot be a vertex. This means that moving from x in direction d does not change the equality constraints: $A(x + \epsilon d) = Ax + \epsilon Ad = Ax = b$.
So, x lies on a line segment in the feasible region and cannot be a vertex.

Basic solutions

Alternative way to find possible vertex solutions:

- Instead of choosing from the restrictions, you choose $n - m$ variables that must be 0.
- These $n - m$ variables are called **non-basic variables**.

A **basis** is a set of linearly independent constraints that determine the coordinates of the vertex.

$$A = \begin{pmatrix} 5 & 2 & 1 & 0 \\ 3 & 3 & 0 & 1 \end{pmatrix} \qquad b = \begin{pmatrix} 180 \\ 135 \end{pmatrix}$$

$$Bx = b \qquad \begin{pmatrix} 5 & 0 \\ 3 & 1 \end{pmatrix} * \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 180 \\ 135 \end{pmatrix}$$

Each basis B is associated with a unique basic solution x.

Basic solutions

If we find $n - m$ non-basic variables and their values are 0, we are left with m variables with m equations.

- The non-basic variables must be chosen so that the rest of the system has a unique solution.
- This solution is called a **basic solution**.

A basic solution that satisfies all non-negativity constraints is called a **basic feasible solution (BFS)**.

The basic feasible solution corresponds to a vertex of the feasible region.

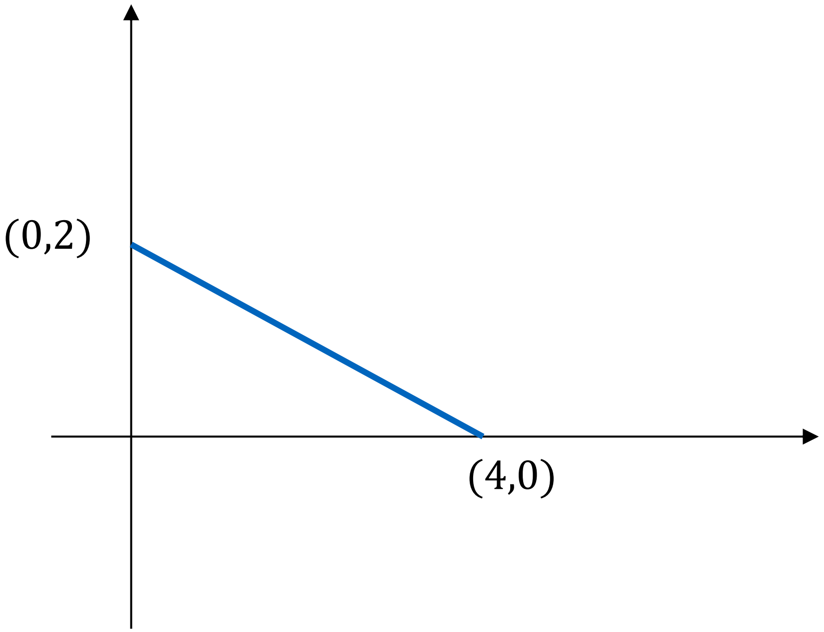
The chosen m variables are called **basic variables** (or **basis**).

Example

max $\quad x_1$

s.t. $\quad x_1 + 2x_2 = 4$

$\qquad x_1, x_2 \geq 0$



$m = 1, n = 2$

There are only 2 vertex solutions: $(0,2)$ and $(4,0)$

$(0,2)$ lies at the intersection of constraints x_1 + 2x_2 = 4 and $x_1 \geq 0$
$(4,0)$ lies at the intersection of constraints x_1 + 2x_2 = 4 and $x_2 \geq 0$
In every solution $n - m = 2 - 1 = 1$ variables equal 0.

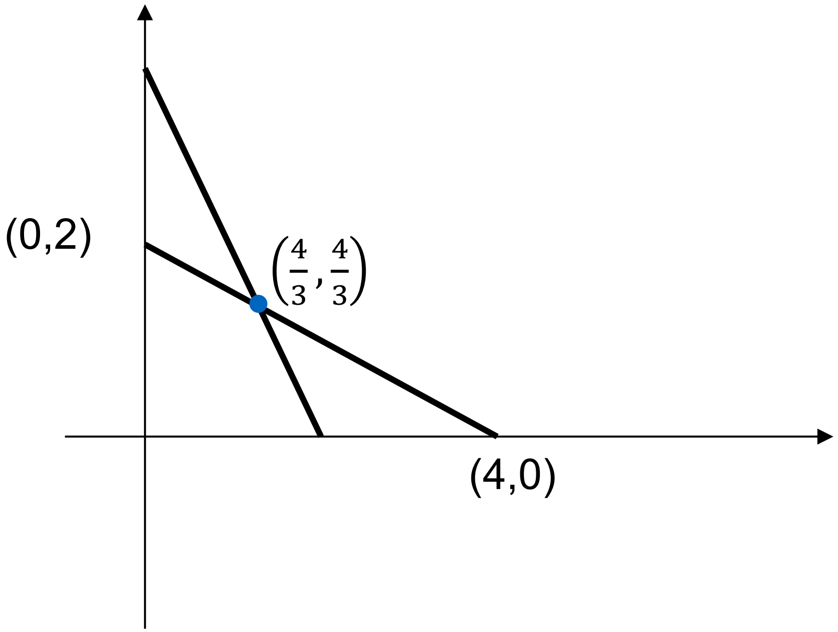
Example

max $\quad x_1$

s.t. $\quad x_1 + 2x_2 = 4$

$\quad 2x_1 + x_2 = 4$

$\quad x_1, x_2 \geq 0$



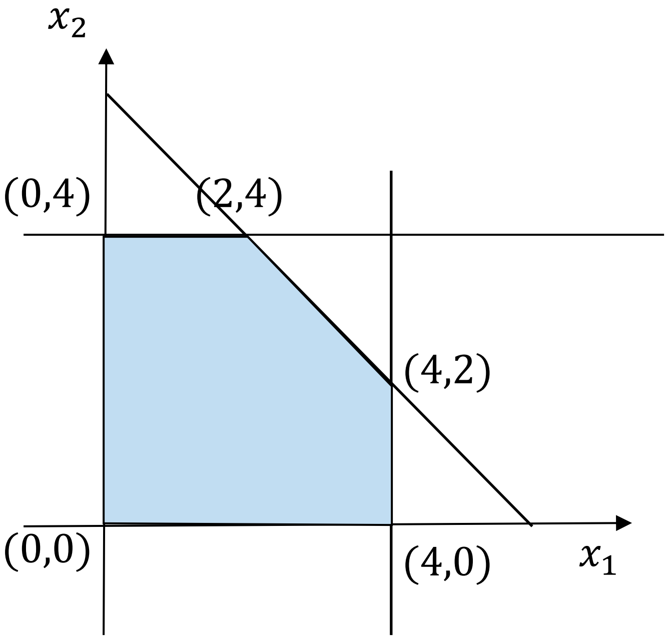
$m = 2, n = 2$

There is only 1 possible vertex solution: $\left(\frac{4}{3}, \frac{4}{3}\right)$

$\left(\frac{4}{3}, \frac{4}{3}\right)$ lies at the intersection of constraints $x_1 + 2x_2 = 4$ and $2x_1 + x_2 = 4$
There are $n - m = 2 - 2 = 0$ variables that equal 0.

Example

$$\begin{array}{lllllll}
\max & 2x_1 & + & x_2 & & & \\
\text{s.t.} & x_1 & + & x_2 & +x_3 & & = 6 \\
& x_1 & & & & +x_4 & = 4 \\
& & & x_2 & & +x_5 & = 4 \\
& & & & & x_1, x_2, x_3, x_4, x_5 & \geq 0
\end{array}$$



$m = 3, n = 5$

Choose $5 - 3 = 2$ variables as non-basic variables.

Example

$$
\begin{aligned}
x_1 \quad + x_2 \quad + x_3 \qquad\qquad\qquad &= 6 \\
x_1 \qquad\qquad\qquad + x_4 \qquad &= 4 \\
x_2 \qquad\qquad\qquad + x_5 &= 4
\end{aligned}
$$

Non-basic	Basis	Solution	BFS
x_1, x_2	$x_3 = 6, x_4 = 4, x_5 = 4$	(0, 0, 6, 4, 4)	yes
x_1, x_3	$x_2 = 6, x_4 = 4, x_5 = -2$	(0, 6, 0, 4, -2)	no **(violates >=0)**
x_1, x_4	impossible, because $x_1+x_4 = 4$ is violated		
x_1, x_5	$x_2 = 4, x_3 = 2, x_4 = 4$	(0, 4, 2, 4, 0)	yes
x_2, x_3	$x_1 = 6, x_4 = -2, x_5 = 4$	(6, 0, 0, -2, 4)	no
x_2, x_4	$x_1 = 4, x_3 = 2, x_5 = 4$	(4, 0, 2, 0, 4)	yes
x_2, x_5	impossible, because $x_2+x_5 = 4$ is violated		
x_3, x_4	$x_1 = 4, x_2 = 2, x_5 = 2$	(4, 2, 0, 0, 2)	yes
x_3, x_5	$x_1 = 2, x_2 = 4, x_4 = 2$	(2, 4, 0, 2, 0)	yes
x_4, x_5	$x_1 = 4, x_2 = 4, x_3 = -2$	(4, 4, -2, 0, 0)	no

How to find a (first) BFS? → *next unit*

Implementation issues

- How to find the (feasible) vertices?
- **How to identify adjacent vertices?**
  - Adjacent vertices share n-1 constraints.
  - Thus, we can enumerate all n constraints to compute all adjacent vertices that share at least n-1 constraints with the currently considered vertex.

Adjacent BFS

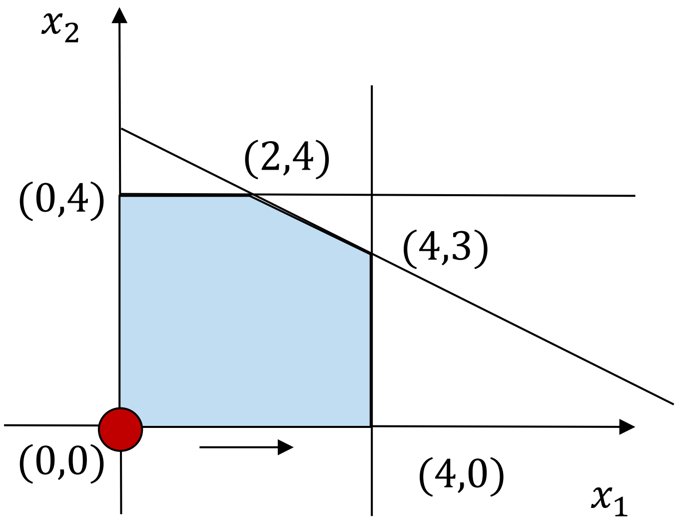
2 vertices for an n variables problem are adjacent, if they share $n - 1$ constraints.

Every vertex must lie at the intersection of m constraints of $Ax = b$.

2 vertices are adjacent if they share $(n - m) - 1$ non-basic variables or share $m - 1$ basic variables.

Example

$$
\begin{array}{llllll}
\max & \overbrace{2x_1 + x_2}^{z :=} & & & & \\
\text{s.t.} & x_1 & + & 2x_2 & \leq & 10 \\
& x_1 & & & \leq & 4 \\
& & & x_2 & \leq & 4 \\
& & & x_1, x_2 & \geq & 0
\end{array}
$$



We introduce a variable z to represent the objective value of the objective function.

$$
\begin{array}{lllllllll}
\max & z & & & & & & & \\
& z & - & 2x_1 & - & x_2 & & = & 0 \\
& & & x_1 & + & 2x_2 & +x_3 & = & 10 \\
& & & x_1 & & & +x_4 & = & 4 \\
& & & & & x_2 & +x_5 & = & 4 \\
& & & & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0
\end{array}
$$

If we choose x_1 and x_2 as non-basic variables, the BFS is $(0, 0, 10, 4, 4)$ with $z = 0$.

Example

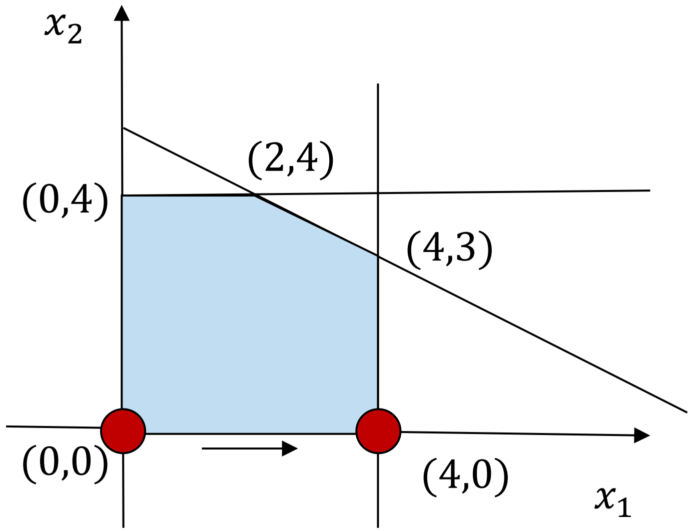
$$
\begin{array}{llcccccccl}
\max & z & & & & & & & & \\
 & z & - & 2x_1 & - & x_2 & & & & = 0 \\
 & & & x_1 & + & 2x_2 & +x_3 & & & = 10 \\
 & & & x_1 & & & & +x_4 & & = 4 \\
 & & & & & x_2 & & & +x_5 & = 4 \\
 & & & & & & & & x_1, x_2, x_3, x_4, x_5 & \geq 0
\end{array}
$$

- Increasing the values x_1 and x_2 will also increase z.
- Because x_1 influences z more, we increase the value of x_1 first.
- What can be changed in the current solution such that the LP is still feasible?
- For each increase in the value of x_1, both x_3 and x_4 must be reduced by one unit.
- How much can we increase the value of x_1?

  If $x_1 = 4$, then $x_4 = 0$. Any further increase makes x_4 negative.

Example

$$
\begin{array}{lllllllll}
\max & z & & & & & & & \\
 & z & - & 2x_1 & - & x_2 & & = & 0 \\
 & & & x_1 & + & 2x_2 & +x_3 & = & 10 \\
 & & & x_1 & & & +x_4 & = & 4 \\
 & & & & & x_2 & +x_5 & = & 4 \\
 & & & & & & x_1, x_2, x_3, x_4, x_5 & \geq & 0
\end{array}
$$



If we set x_1 to 4, we get $x_4 = 0$.

⇒ New non-basic variables: x_2 and x_4.

The new solution is $(4, 0, 6, 0, 4)$ with $z = 8$

Summary

The example shows an iteration of the Simplex algorithm.

In the example, x_1 is the entering basic variable and x_4 is the exiting basic variable.

The values of the original BFS were easy to calculate because all basic variables x_3, x_4, x_5 appeared in a single equation in which the non-basic variables x_1, x_2, were set to 0.

But how do we calculate the new basic variables if this is not the case?

Canonical form

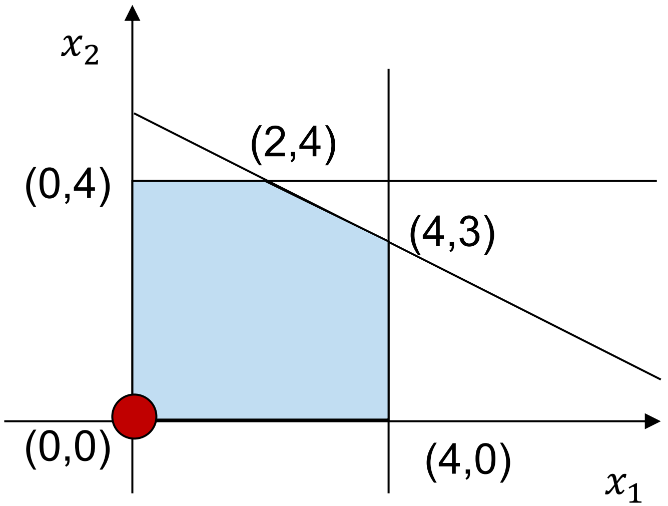
- Canonical form of a linear program:
  Linear system of equations (standard form)
  + Each equation contains a variable with coefficient 1, which has coefficient 0 in all other equations.

- After each Simplex iteration, the system of equations should be converted into canonical form, where the basic variables occur in only one equation with coefficient 1.
  → In canonical form, it is easy to calculate how the basic variables and z change when a non-basic variable is increased.

- The transformation into canonical form can be done using row operations.

Example

$$
\begin{array}{llllllll}
z & -2x_1 & -x_2 & & & & = 0 & (0) \\
& x_1 & +2x_2 & +x_3 & & & = 10 & (1) \\
& x_1 & & & +x_4 & & = 4 & (2) \\
& & x_2 & & & +x_5 & = 4 & (3) \\
& & & & & x_1, \dots, x_5 \geq 0 & &
\end{array}
$$

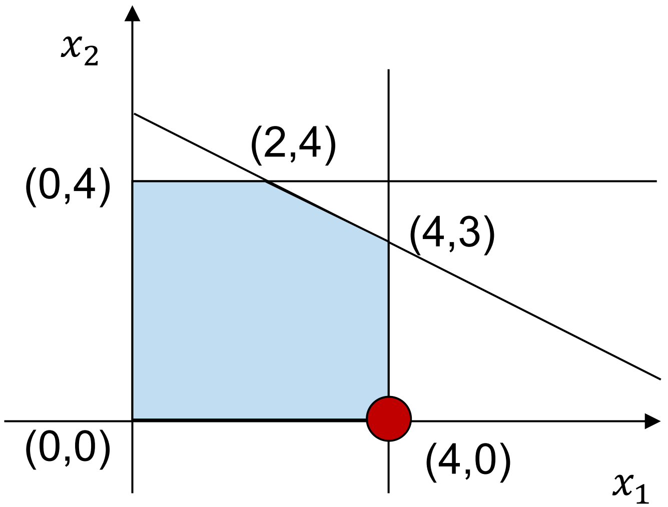


Operations to get basic variables x_1, x_3, x_5:

Add $2 \cdot (2)$ to (0)

Add $-1 \cdot (2)$ to (1)

$$
\begin{array}{llllllll}
z & & -x_2 & & +2x_4 & & = 8 & (0) \\
& & 2x_2 & +x_3 & -x_4 & & = 6 & (1) \\
& x_1 & & & +x_4 & & = 4 & (2) \\
& & x_2 & & & +x_5 & = 4 & (3) \\
& & & & & x_1, \dots, x_5 \geq 0 & &
\end{array}
$$



Example

$$
\begin{array}{lllllll}
z & -x_2 & & +2x_4 & & = 8 & \text{(0)} \\
& 2x_2 & +x_3 & -x_4 & & = 6 & \text{(1)} \\
x_1 & & & +x_4 & & = 4 & \text{(2)} \\
& x_2 & & & +x_5 & = 4 & \text{(3)} \\
& & & & x_1, \dots, x_5 & \geq 0 &
\end{array}
$$

New basic solution: $(4{,}0{,}6{,}0{,}4)$
The coefficients of each variable in equation (0) show how an increase affects z.

In this case, an increase in the value of x_2 would increase z.

How much can we increase the values of x_2 in the current basic solution $(4{,}0{,}6{,}0{,}4)$ by changing the values of other variables?

Observations

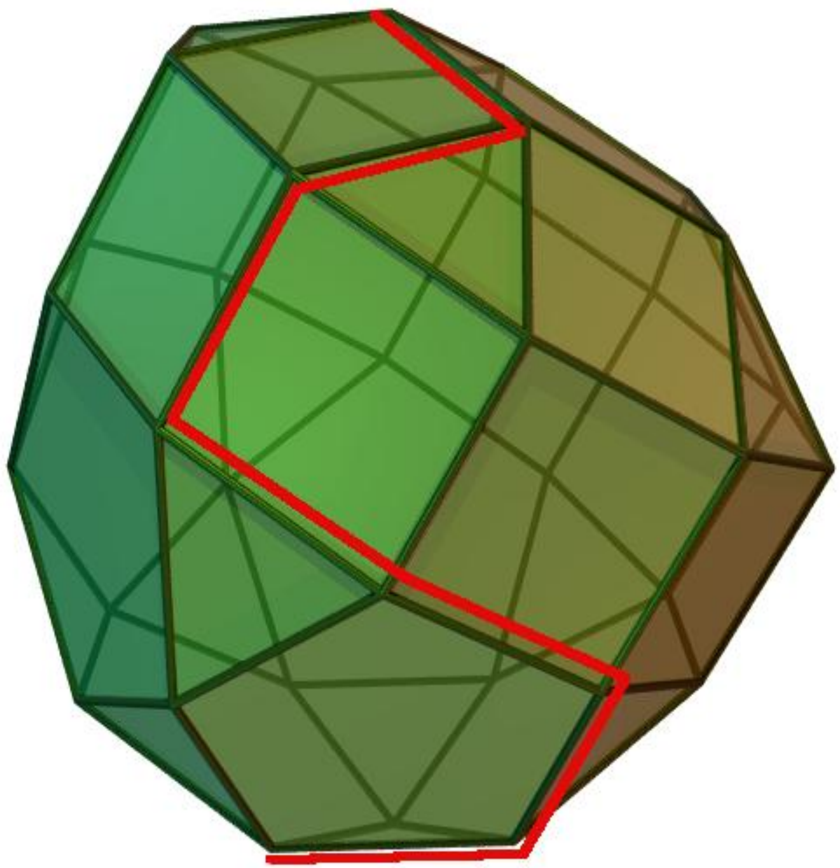
The Simplex method was performed without enumerating all possible adjacent BFSs in each iteration.

A non-basic variable was identified in each iteration, such that an increase in its value led to an increase in z.

We changed the values of the non-basic and basic variables, such that all equations were satisfied.

In all cases, the value of a selected non-basic variable was increased and at least one other non-basic variable was decreased and eventually set to 0.

Simplex algorithm



Source: Wikipedia

Summary

Linear programs describe convex polytopes.

The solutions of LPs are on the vertices of their polytopes.

Vertices are obtained at the intersection of constraints.

Vertices are basic feasible solutions of the associated linear system of equations in standard form.

Due to convexity, optimality can be determined by looking at adjacent BFSs.

Simplex algorithm finds adjacent BFSs.
(More in the next unit)